Advanced Design Scripting & Parametrics – Class 07 – Meshes and BREPs

Today – key concepts

- Meshes
- Boundary Representations (BREPS)
- Surface curvature if we have time

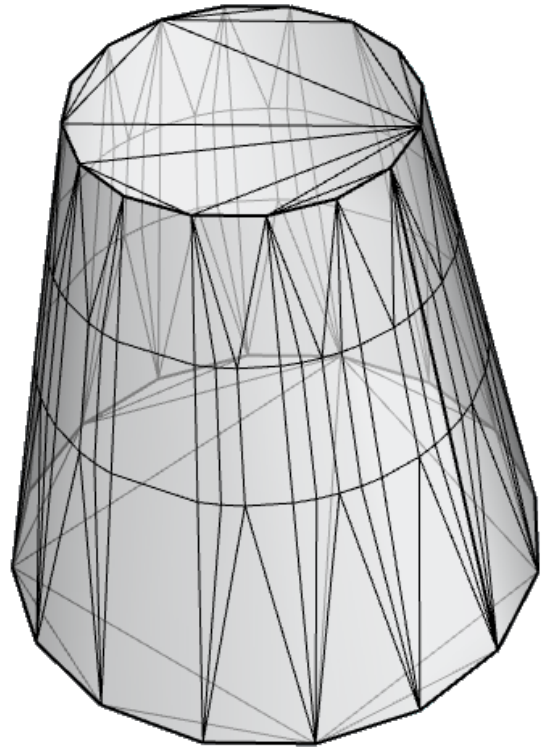
Meshes

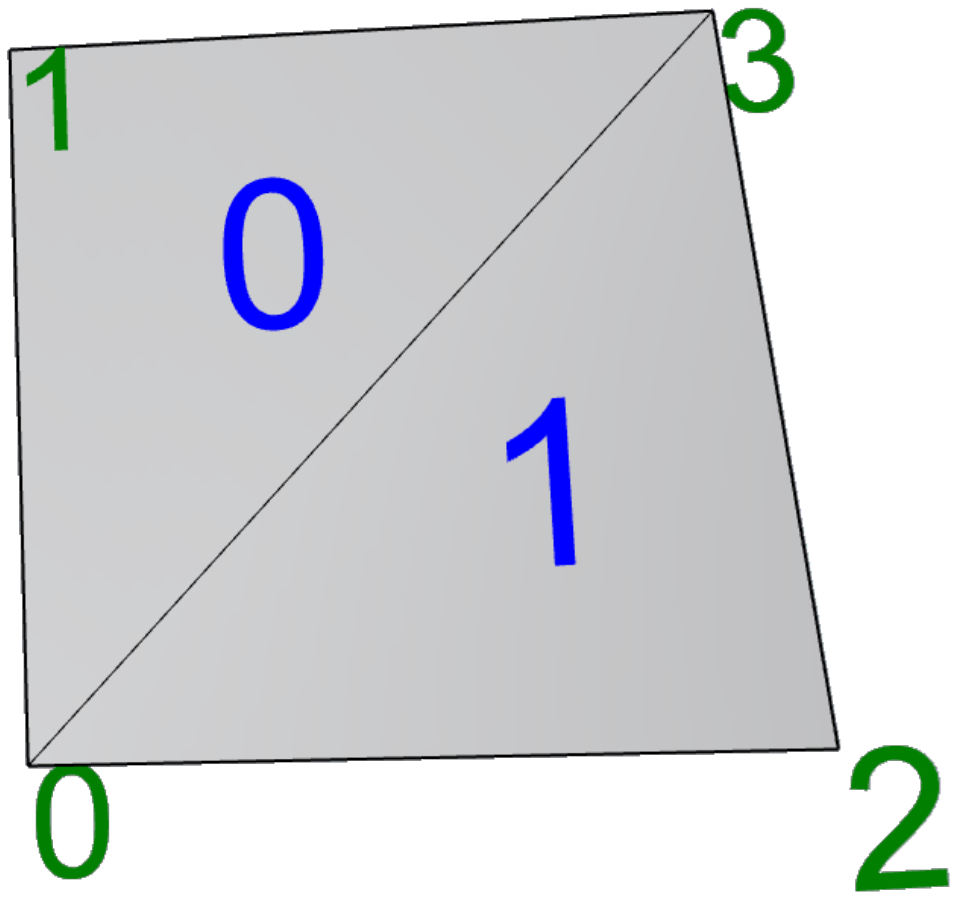
Meshes are collections of triangles (and sometimes “quads”), assembled into contiguous closed or open configurations.

Several triangles can share a common vertex, so there are efficient ways of making these references between triangles and vertices.

Vertices can have additional information, including:

- Color
- Effective normal
- Positioning in 2D texture space



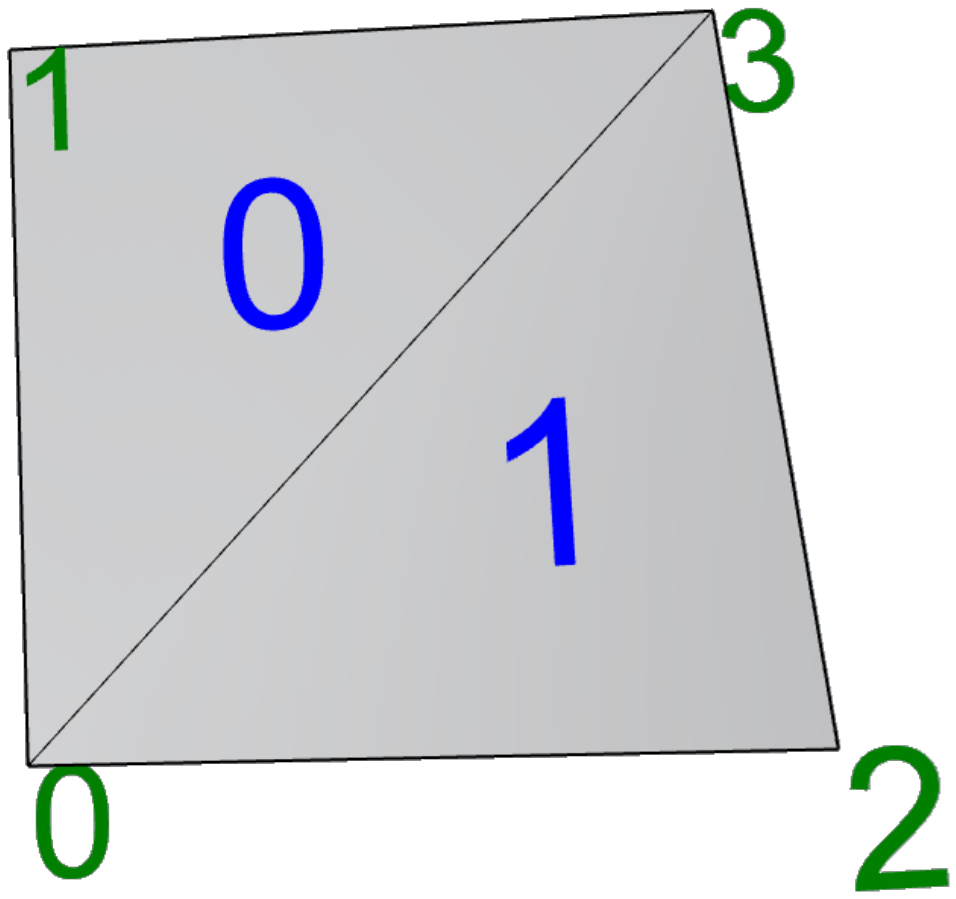


Mesh elements

Comprised of:

- **Vertices** – each has:
  - an ***index*** (integer) corresponding to its index in the vertex array

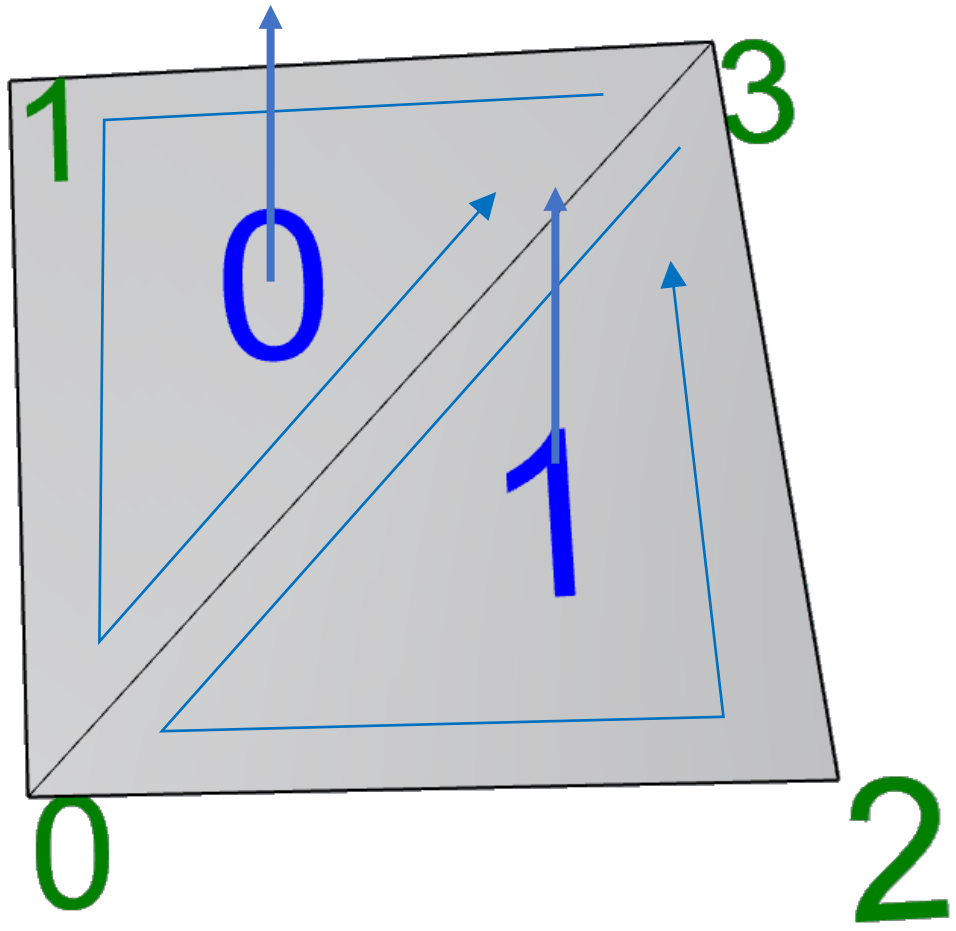
  and optionally:
  - A ***normal*** (Vector3f)
  - A ***texture coordinate*** (Point2f)
  - A ***color*** (Color)
- **Faces** - each has
  - 4 vertex ***indices*** A, B, C, D (integers), where the C and D can be duplicates can be duplicates
  - An implied surface ***normal*** (Vector3f), with a direction “pointing out” defined by the cross product of the vectors AB and BC
- **Edges** – implied, which can be:
  - Internal
  - “Naked”
  - Non-manifold



Mesh elements

A mesh can be:

- Open
- Closed
- Non-manifold (almost never is)

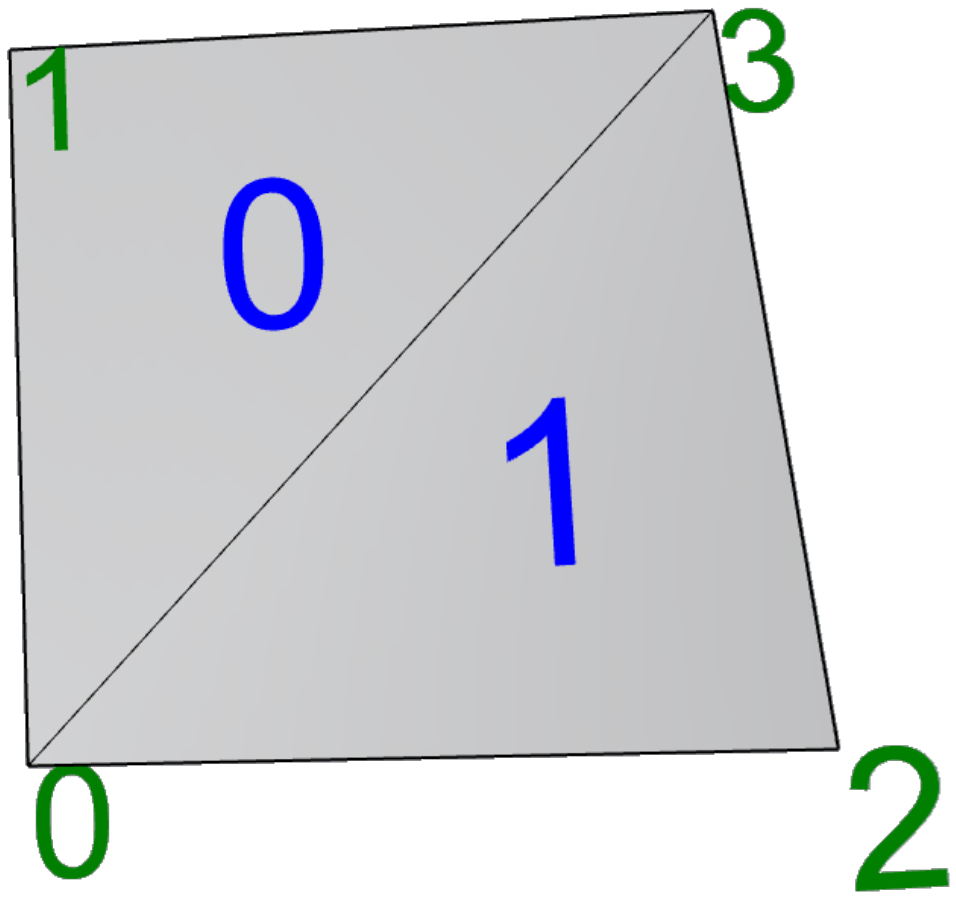


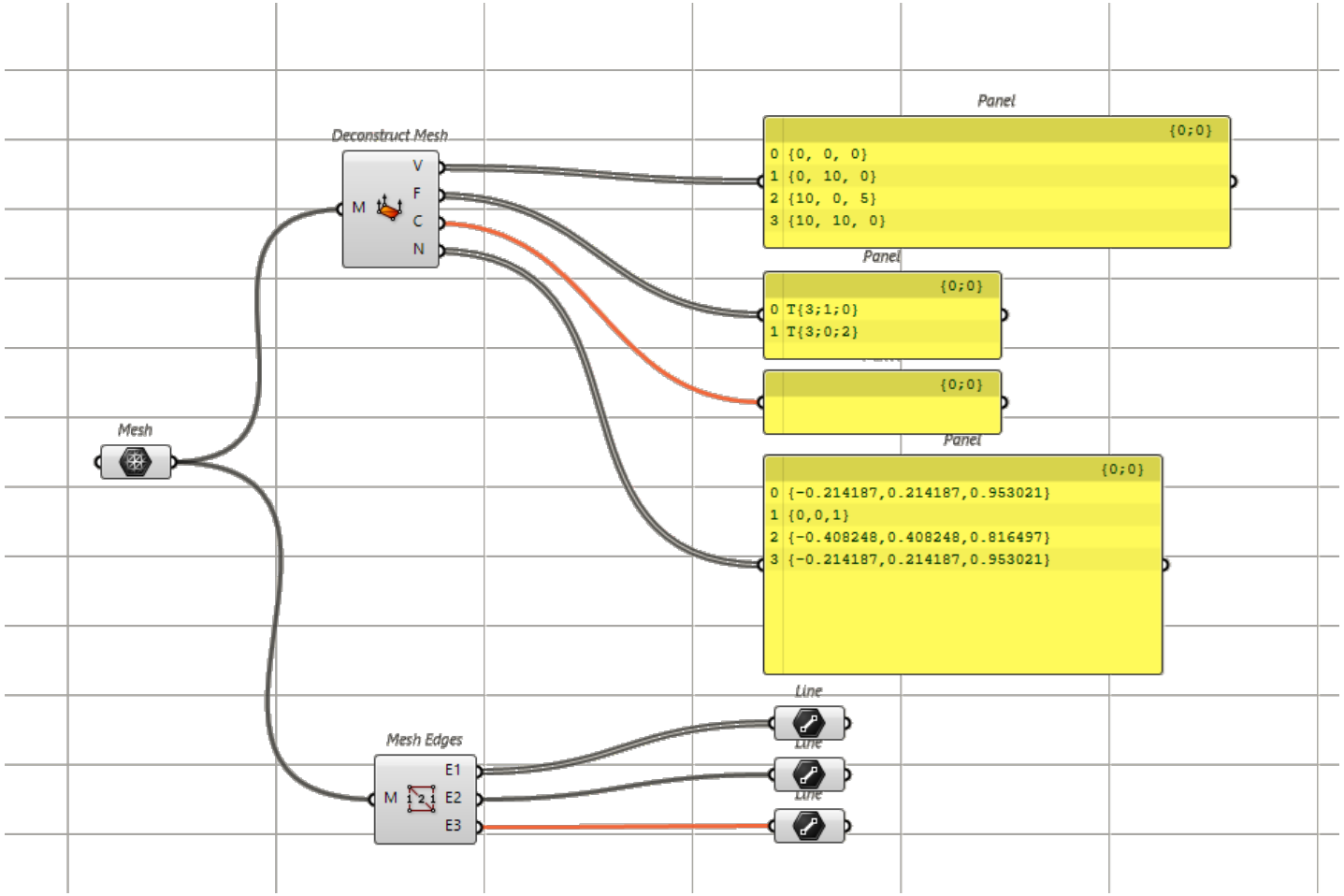
AnalyzeMesh.py

vertices	4 Item(s)	list
[0]	0,0,0	Point3f
[1]	0,10,0	Point3f
[2]	10,0,5	Point3f
[3]	10,10,0	Point3f
faces	2 Item(s)	MeshFace
[0]	T(3, 1, 0)	MeshFace
A	3	int
B	1	int
C	0	int
D	0	int
IsQuad	False	bool
IsTriangle	True	bool
Unset	T(-2147483648, -2	MeshFace
[1]	T(3, 0, 2)	MeshFace
A	3	int
B	0	int
C	2	int
D	2	int
IsQuad	False	bool
IsTriangle	True	bool
Unset	T(-2147483648, -2	MeshFace
Capacity	2	int
Count	2	int
QuadCount	0	int
TriangleCou	2	int

normals	2 Item(s)	MeshFace
[0]	0,0,1	Vector3f
[1]	-0.4082483,0.4082483,0.8164966	Vector3f
Capacity	2	int
Count	2	int
textureCoords	4 Item(s)	MeshText
[0]	0,0	Point2f
[1]	0,1	Point2f
[2]	1,0	Point2f
[3]	1,1	Point2f

AnalyzeMesh.gh





- Mesh
  - Constructors
  - Properties
    - ComponentStates
    - DisjointMeshCount
    - Disposed
    - FaceNormals
    - Faces
    - HasBrepForm
    - HasCachedTextureCoordinates
    - HasPrincipalCurvatures
    - HasUserData
    - IsClosed
    - IsDeformable
    - IsDocumentControlled
    - IsDocumentControlled
    - IsOriented
    - IsSolid
    - IsValid
    - Ngons
    - Normals
    - ObjectType
    - PartitionCount
    - PerformCorruptionTesting
    - TextureCoordinates
    - TopologyEdges
    - TopologyVertices
    - UserData
    - UserDictionary
    - UserStringCount
    - VertexColors
    - Vertices
  - Methods
    - Append
    - Check
    - ClearSurfaceData
    - ClearTextureData
    - ClosestMeshPoint
    - ClosestPoint
    - CollapseFacesByArea
    - CollapseFacesByByAspectRatio
    - CollapseFacesByEdgeLength
    - ColorAt
    - Compact
    - ComponentIndex
    - ComputeAutoCreaseInformation
    - ComputeCurvatureApproximation
    - ComputeThickness
    - ConstructConstObject
    - CopyFrom
    - CreateBooleanDifference
    - CreateBooleanIntersection
    - CreateBooleanSplit
    - CreateBooleanUnion
    - CreateContourCurves
    - CreateConvexHull3D
    - CreateExtrusion
    - CreateFromBox
    - CreateFromBrep
    - CreateFromClosedPolyline
    - CreateFromCone
    - CreateFromCurveExtrusion
    - CreateFromCurvePipe
    - CreateFromCylinder
    - CreateFromExtrusion
    - CreateFromFilteredFaceList
    - CreateFromIterativeCleanup
    - CreateFromLines
    - CreateFromPatchSingleFace
    - CreateFromPlanarBoundary
    - CreateFromPlane
    - CreateFromSphere
    - CreateFromSubD
    - CreateFromSubDControlNet
    - CreateFromSubDControlNetWithTextureCoordinat
    - CreateFromSurface
    - CreateFromSurfaceControlNet
    - CreateFromTessellation
    - CreateFromTorus
    - CreateIcoSphere
    - CreatePartitions
    - CreatePatch
    - CreateQuadSphere
    - CreateRefinedCatmullClarkMesh
    - CreateRefinedLoopMesh
    - CreateUnweldedMesh
    - CreateVertexColorsFromBitmap
    - DataCRC
    - DeleteAllUserStrings
    - DeleteUserString
    - DestroyPartition
    - DestroyTopology
    - DestroyTree
    - Dispose
    - Dispose
    - Duplicate
    - Duplicate
    - DuplicateMesh
    - DuplicateShallow
    - EnsurePrivateCopy
    - EvaluateMeshGeometry
    - ExplodeAtUnweldedEdges
    - ExtendSelectionByEdgeRidge
    - ExtendSelectionByFaceLoop
    - ExtractNonManifoldEdges
    - FileHole
    - FillHoles
    - Flip
    - FromBase64String
    - FromJSON
    - GeometryEquals
    - GeometryReferenceEquals
    - GetBoundingBox
    - GetCachedTextureCoordinates
    - GetNakedEdgePointStatus
    - GetNakedEdges
    - GetNgonAndFacesCount
    - GetNgonAndFacesEnumerable
    - GetObjectData
    - GetOutlines
    - GetPartition
    - GetSelfIntersections
    - GetUnsafeLock
    - GetUserString
    - GetUserStrings
    - HealNakedEdges
    - InvalidateCachedTextureCoordinates
    - IsManifold
    - IsPointInside
    - IsValidWithLog
    - MakeDeformable
    - MatchEdges
    - MemoryEstimate
    - MergeAllCoplanarFaces
    - NonConstOperation
    - NonConstOperation
    - NonConstOperation
    - NormalAt
      - (int, double, double, double, double)
      - (MeshPoint)
    - Offset
    - OnSwitchToNonConst
    - OnSwitchToNonConst
    - OnSwitchToNonConst
    - PatchSingleFace
    - PointAt
    - PullCurve
    - PullPointsToMesh
    - QuadRemesh
    - QuadRemeshAsync
    - QuadRemeshBrep
    - QuadRemeshBrepAsync
    - RebuildNormals
    - Reduce
    - ReleaseUnsafeLock
    - RequireIterativeCleanup
    - Rotate
    - Scale
    - SetCachedTextureCoordinates
    - SetCachedTextureCoordinatesFromMaterial
    - SetTextureCoordinates
    - SetUserString
    - ShrinkWrap
    - Smooth
    - SolidOrientation
    - Split
    - SplitDisjointPieces
    - SplitWithProjectedPolylines
    - Subdivide
    - ToJSON
    - Transform
    - Translate
    - UnifyNormals
    - Unweld
    - UnweldEdge
    - UnweldVertices
    - Volume
    - Weld
    - WithDisplacement
    - WithEdgeSoftening
    - WithShutLining

Mesh Geometries

- Mesh
- MeshBooleanOptions
- MeshCheckParameters
- MeshDisplacementInfo
- MeshExtruder
- MeshExtruderFaceDirectionMode
- MeshExtruderParameterMode
- MeshFace
- MeshingParameters
- MeshingParameterStyle
- MeshingParameterTextureRange
- MeshNgon
- MeshPart
- MeshPipeCapStyle
- MeshPoint
- MeshThicknessMeasurement
- MeshType
- MeshUnsafeLock
- MeshUnwrapMethod
- MeshUnwrapper

MeshFace

- Constructors
- Properties
  - A
  - B
  - C
  - D
  - IsQuad
  - IsTriangle
  - Item
  - Unset
- Methods
  - CompareTo
  - Equals
  - Flip
  - GetHashCode
  - IsValid
  - IsValidEx
  - Repair
  - RepairEx
  - Set
  - ToString
- Operators
  - !=
  - ==

Mesh Collections

- MeshFaceList
- MeshFaceNormalList
- MeshNgonList
- MeshTextureCoordinateList
- MeshTopologyEdgeList
- MeshTopologyVertexList
- MeshVertexColorList
- MeshVertexList
- MeshVertexNormalList
- MeshVertexStatusList

MeshVertexList

- Properties
  - Capacity
  - Count
  - Item — Point3f
  - UseDoublePrecisionVertices
- Methods
  - Add
  - AddVertices
  - Align
  - Clear
  - CombineIdentical
  - CullUnused
  - Destroy
  - GetConnectedVertices
  - GetEnumerator
  - GetTopologicalIndenticalVertices
  - GetVertexFaces
  - Hide
  - HideAll
  - IsHidden
  - Point3dAt
  - Remove
  - SetVertex
  - Show
  - ShowAll
  - ToFloatArray
  - ToPoint3dArray
  - ToPoint3fArray

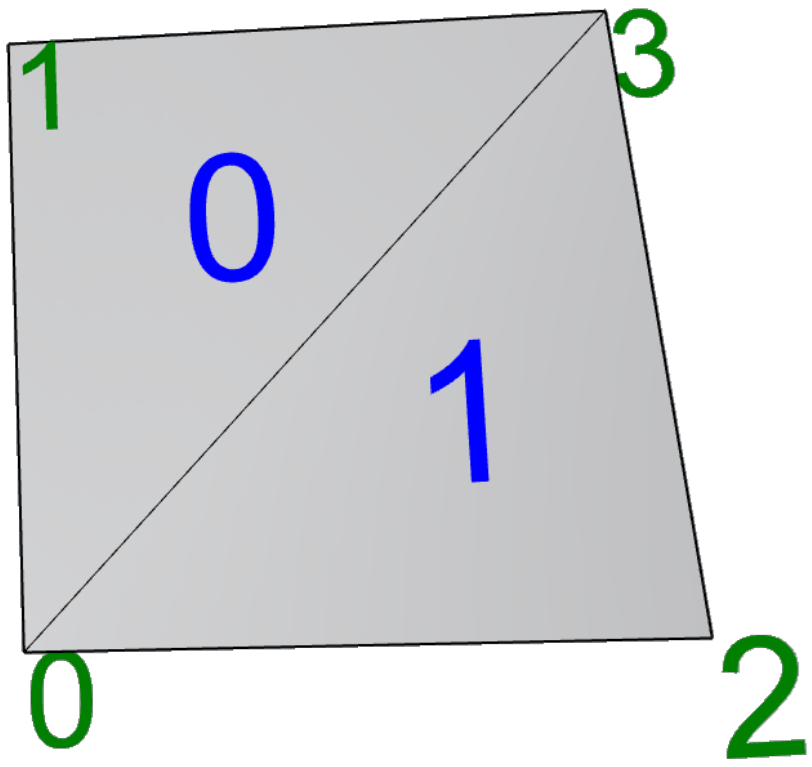
MeshVertexColorList

- Properties
  - Capacity
  - Count
  - Item — Color
  - Tag
- Methods
  - Add
  - AddRange
  - AppendColors
  - Clear
  - CreateMonotoneMesh
  - Destroy
  - GetEnumerator
  - SetColor
  - SetColors
  - ToARGBArray

MeshVertexNormalList

- Properties
  - Capacity
  - Count
  - Item — Vector3f
- Methods
  - Add
  - AddRange
  - Clear
  - ComputeNormals
  - Destroy
  - Flip
  - GetEnumerator
  - SetNormal
  - SetNormals
  - ToFloatArray
  - UnitizeNormals

OBJ Definition (“1 Ordered”)



.OBJ file

```
1  # Rhino
2
3  mtllib smallmesh.mtl
4  usemtl diffuse_0_0_0_255
5  v 0 0 0
6  v 0 10 0
7  v 10 0 5
8  v 10 10 0
9  vt 0 0
10 vt 0 1
11 vt 1 0
12 vt 1 1
13 vn -0.214 0.214 0.953
14 vn 0 0 1
15 vn -0.408 0.408 0.816
16 vn -0.214 0.214 0.953
17 f 4/4/4 2/2/2 1/1/1
18 f 4/4/4 1/1/1 3/3/3
```

vertices

texture

normals

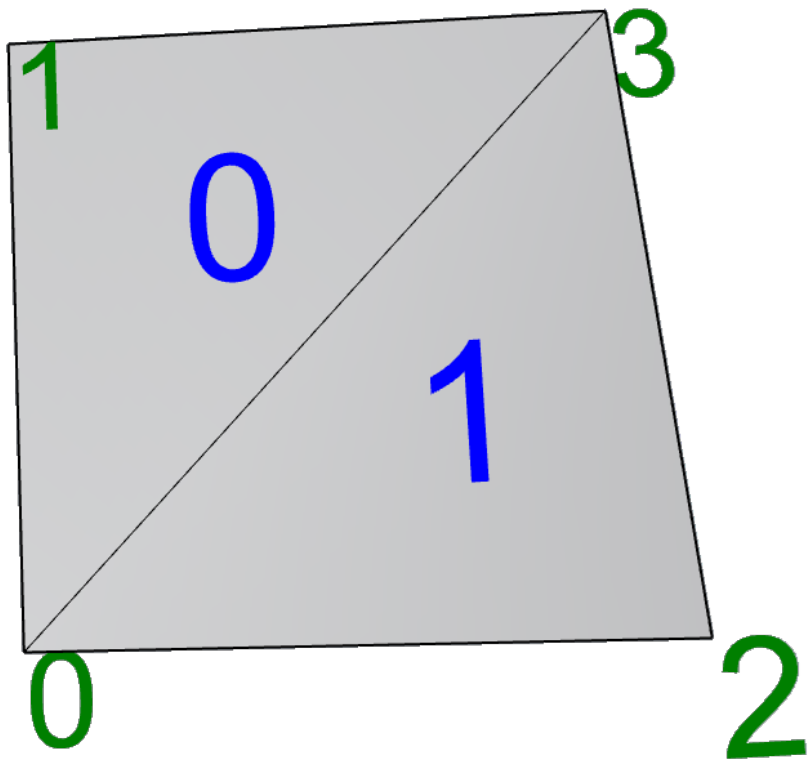
faces

v t n

.MTL file

```
1  # Rhino
2  newmtl diffuse_0_0_0_255
3  Ka 0.0000 0.0000 0.0000
4  Kd 0.0000 0.0000 0.0000
5  Ks 1.0000 1.0000 1.0000
6  Tf 1.0000 1.0000 1.0000
7  d 1.0000
8  Ns 0.0000
9  Ni 1.0000
```

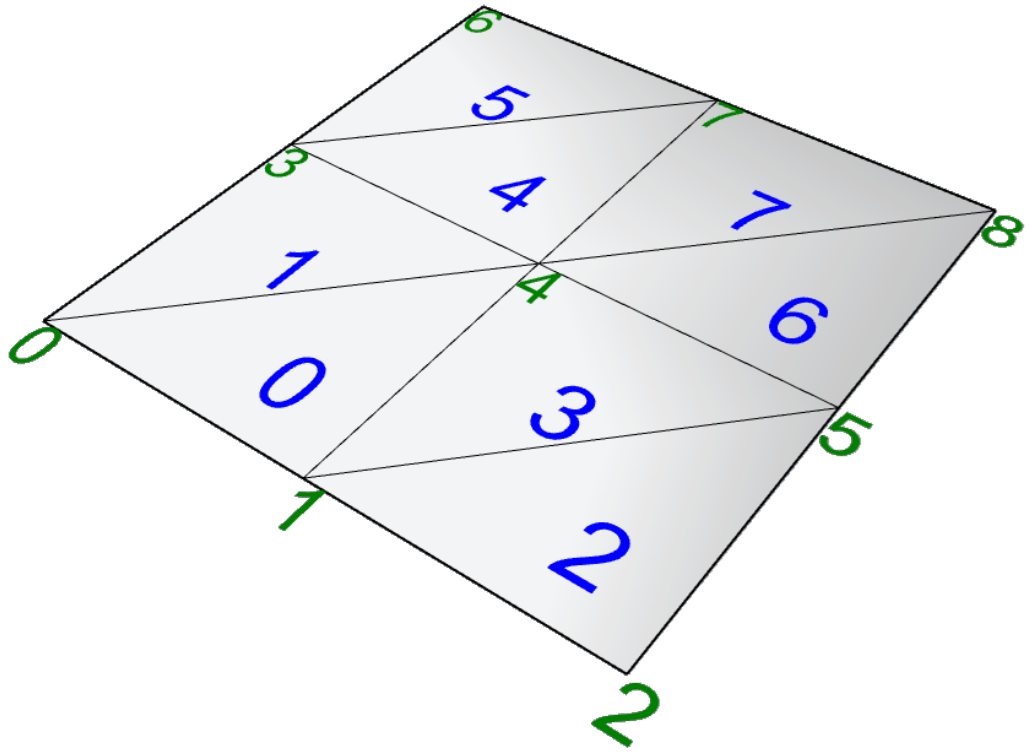
STL Definition



.STL file

```
solid OBJECT
  facet normal 0 -0 1
    outer loop
      vertex 10 10 0
      vertex 0 10 0
      vertex 0 0 0
    endloop
  endfacet
  facet normal -0.408 0.408 0.816
    outer loop
      vertex 10 10 0
      vertex 0 0 0
      vertex 10 0 5
    endloop
  endfacet
endsolid OBJECT
```

Making a mesh (makeMesh01.gh)



index

ranges

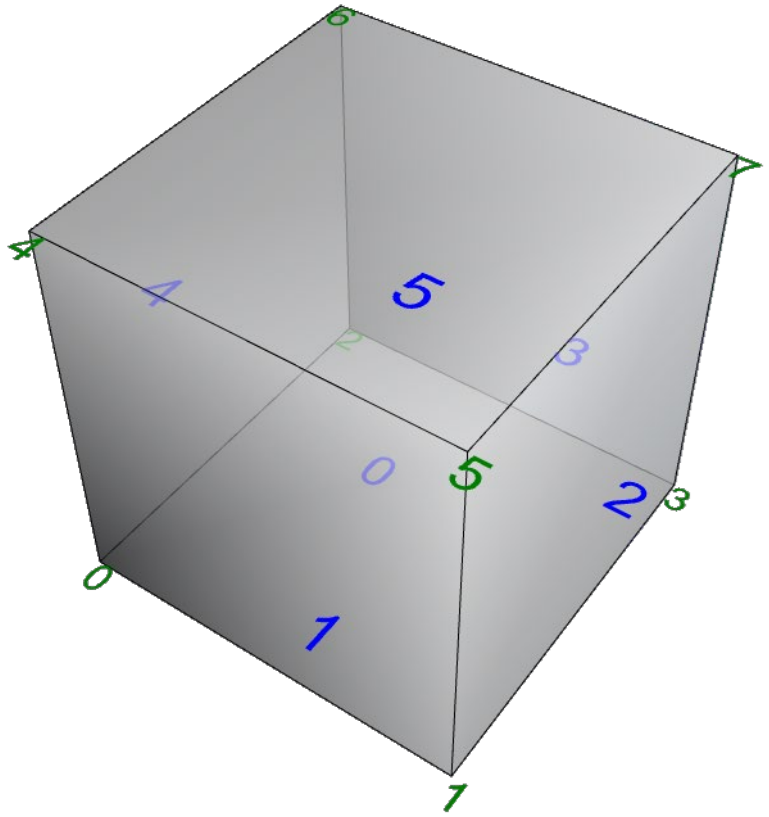
vertices

faces

mesh

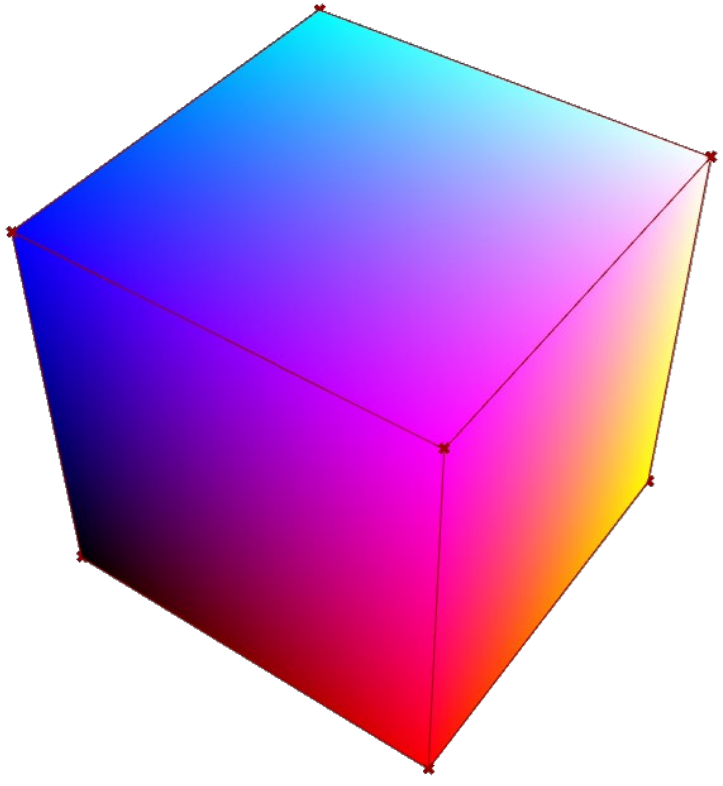
```
11  #index helper functions
12  def VID(i, j):
13      return j * countX + i
14  def V(i, j):
15      return vertices(VID(i, j))
16
17  #make x values
18  xvals = numpy.arange(0, sizeX, sizeX / float(countX - 1))
19  xvals = numpy.append(xvals, sizeX)
20  #print(xvals)
21  countX = xvals.size
22
23  #make y values
24  yvals = numpy.arange(0, sizeY, sizeY / float(countY - 1))
25  yvals = numpy.append(yvals, sizeY)
26  yCount = yvals.size
27
28  #create the vertices that the edges will reference later
29  vertices = []
30  for y in yvals:
31      i = 0
32      for x in xvals:
33          vertices.append((x,y,0.0))
34
35  # make the face vertex lists
36  faceVertices = []
37  j = 0
38  for y in yvals:
39      i = 0
40      for x in xvals:
41          if (i < countX - 1) and (j < countY - 1):
42                  #### forward faces
43                  faceVertices.append((VID(i,j), VID(i+1,j), VID(i+1,j+1)))
44                  faceVertices.append((VID(i,j), VID(i+1,j+1), VID(i,j+1)))
45          i = i + 1
46      j = j + 1
47
48  #Make the mesh
49  mesh = rs.AddMesh( vertices, faceVertices )
```

Making a cube (makeMesh04 – cube)

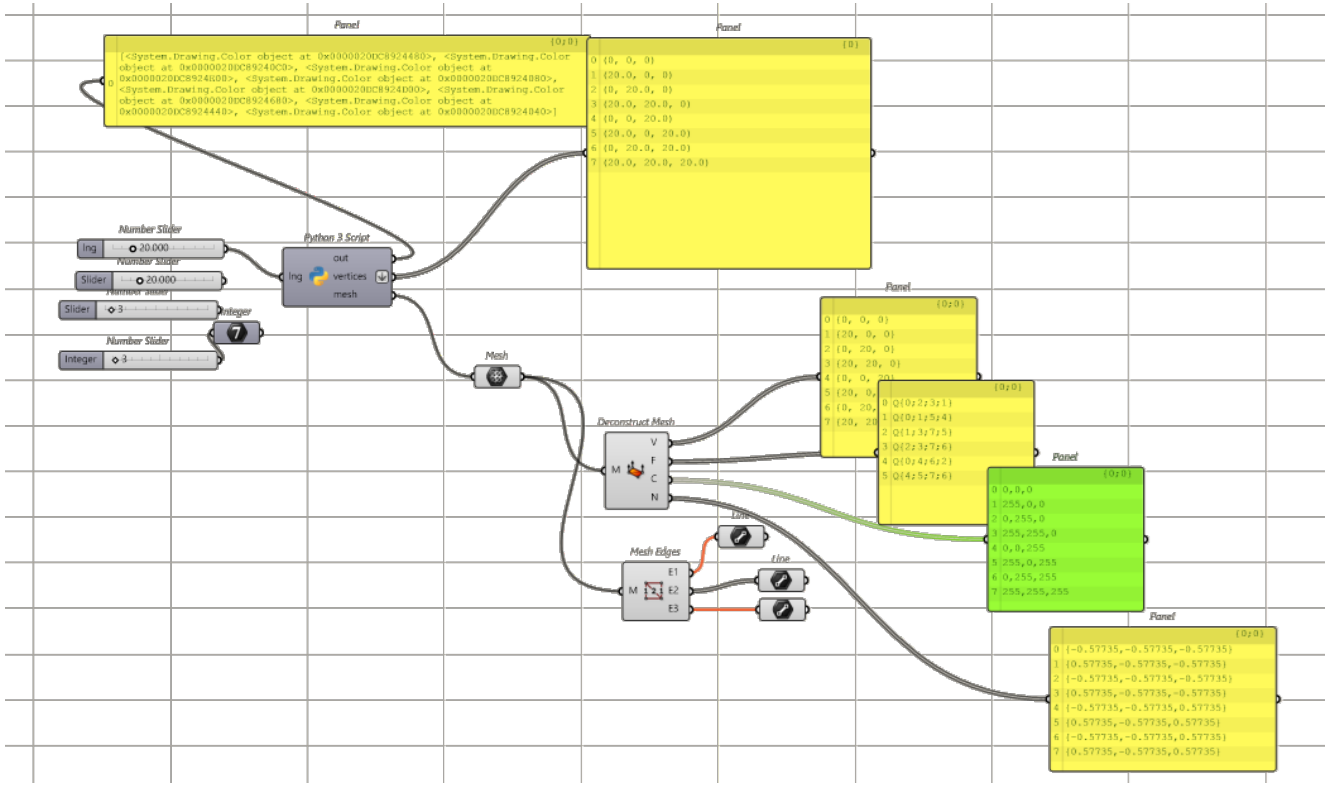



```
12  #create the vertices that the edges will reference later
13  vertices = []
14  vertices.append((0,0,0))
15  vertices.append((lng,0,0))
16  vertices.append((0,lng,0))
17  vertices.append((lng,lng,0))
18
19  vertices.append((0,0,lng))
20  vertices.append((lng,0,lng))
21  vertices.append((0,lng,lng))
22  vertices.append((lng,lng,lng))
23
24
25  faceVertices = []
26
27  faceVertices.append((0,2,3,1)) #bottom
28  faceVertices.append((0,1,5,4)) #front
29  faceVertices.append((1,3,7,5)) #right
30  faceVertices.append((2,3,7,6)) #back
31  faceVertices.append((0,4,6,2)) #left
32  faceVertices.append((4,5,7,6)) #top
33
34
35  #Make the mesh
36  mesh = rs.AddMesh( vertices, faceVertices )
```

Making a cube with colors (makeMesh05 – cube)



```
36  vcolors = []
37  for i in range(8):
38      vcolors.append(rs.CreateColor(int(vertices[i][0] / lng * 255), \
39      int(vertices[i][1] / lng * 255), \
40      int(vertices[i][2] / lng * 255)))
41
42  print(vcolors)
43  #Make the mesh
44  mesh = rs.AddMesh( vertices, faceVertices, vertex_colors = vcolors )
```

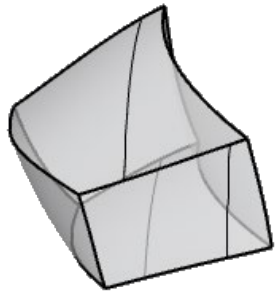


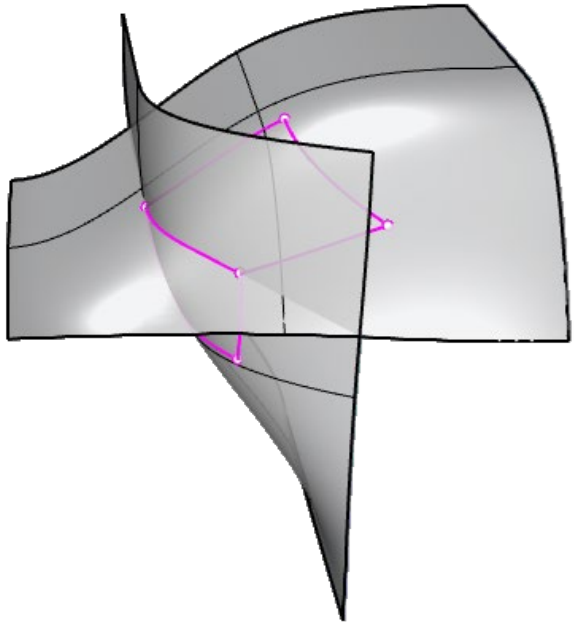
BREPs (Boundary Representations)

BREPs are collections of trimmed or untrimmed NURBS surfaces,

that are connected into open or closed objects.

In Rhino, Solids are closed BREPS





BREPs (Boundary Representations)